Bioinformatyka 2

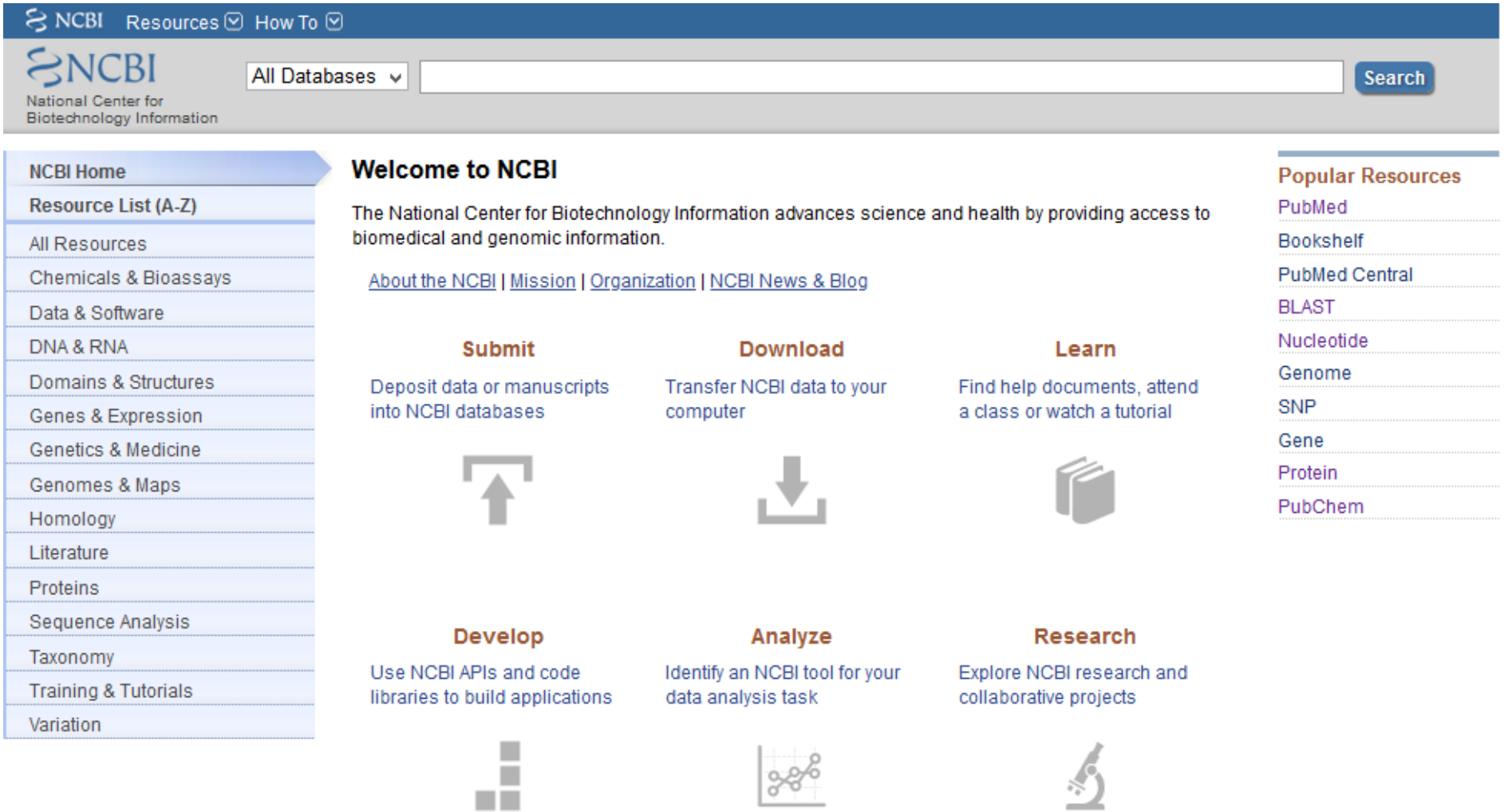
ZBOiB WBBiB UJ
Adrian Kania

27 października 2026

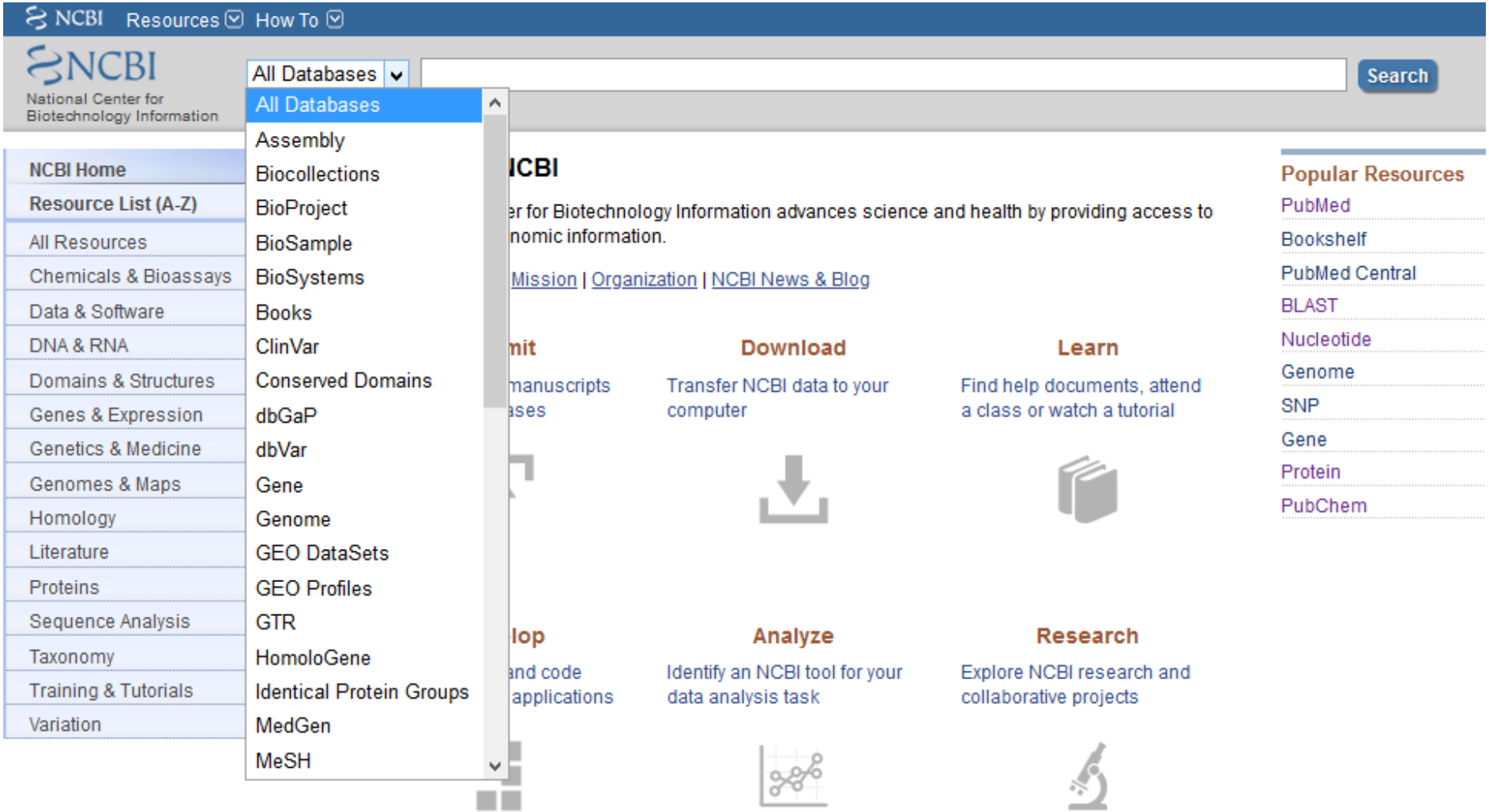
NCBI

1. **NCBI** – National Center for Biotechnology Information
2. **NCBI Entrez** – zintegrowany system ponad 30 baz danych gromadzących informację z nauk biomedycznych, głównie w postaci sekwencji i literatury naukowej. Obejmuje m.in Pubmed (streszczenia publikacji), Nucleotide (sekwencje nukleotydowe), Protein (sekwencje aminokwasowe).
3. **Entrez Programming Utilities** – oprogramowanie umożliwiające dostęp do danych zintegrowanych w ramach systemu Entrez bez konieczności obsługi formularzy na stronach WWW
    1. **ESearch** – wyszukiwanie identyfikatorów/kodów dostępu rekordów wybranej bazy danych,
    2. **EFetch** – pobieranie rekordów o wskazanych identyfikatorach lub wyników wskazanego wyszukiwania

NCBI


NCBI Resources How To
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EFetch

EFetch

EFetch (efetch.fcgi) returns full data records for a list of unique identifiers (UIDs) in a format specified in the parameters. The list of UIDs is either provided in the parameters, or is retrieved from the History server.

EFetch Parameters

EFetch Required Parameters

- `db` (required): Database containing the unique identifiers (UIDs) for which you wish to retrieve records. You can see NCBI's table of Entrez Unique Identifiers (UIDs) for a complete list of allowable database names, but some example values include:
  - `pubmed`: PubMed
  - `pmc`: PubMed Central
  - `nlmcatalog`: NLM Catalog
- `id` (required): Either a single unique identifier (UID) or a comma-delimited list of UIDs. All of the UIDs must be from the database specified by the `db` parameter.

EFetch Optional Parameters

- `retstart` (optional): Setting this parameter helps limit which records will be shown in the output, as it determines whether the record for the first input unique identifier (UID) is retrieved, or whether to skip to a later UID in the input list. For example, if `retstart` is set to `10`, the output will begin with the record for the tenth UID. The default of this parameter is `1`, corresponding to the first UID in the input list. This parameter can be used in conjunction with `retmax` to download an arbitrary subset of records.
- `retmax` (optional): Total number of records to be shown in the output, up to a maximum of 10,000. If the set of records you are trying to retrieve is larger than 10,000, you can submit multiple EFetch requests, and increase the `retstart` parameter each time.
- `retmode`/`rettype`: These two parameters determine how your results will be displayed. `retmode` determines the data format your records will be returned in (e.g. XML, plain text, etc.). `rettype` determines the specific view your records will be returned in (e.g. MEDLINE, Abstract, list of PMIDs, etc.). Different databases have different allowable data formats and record views, and not all `retmode` data formats are compatible with all `rettype` record views, and vice versa.

  The table below shows the allowable combinations of `retmode` and `rettype` for some of the databases. **Bold** `retmode` values are the default data format for the specified database. **Bold** `rettype` parameters are the default record view for the specified data format and database.

EFetch Examples

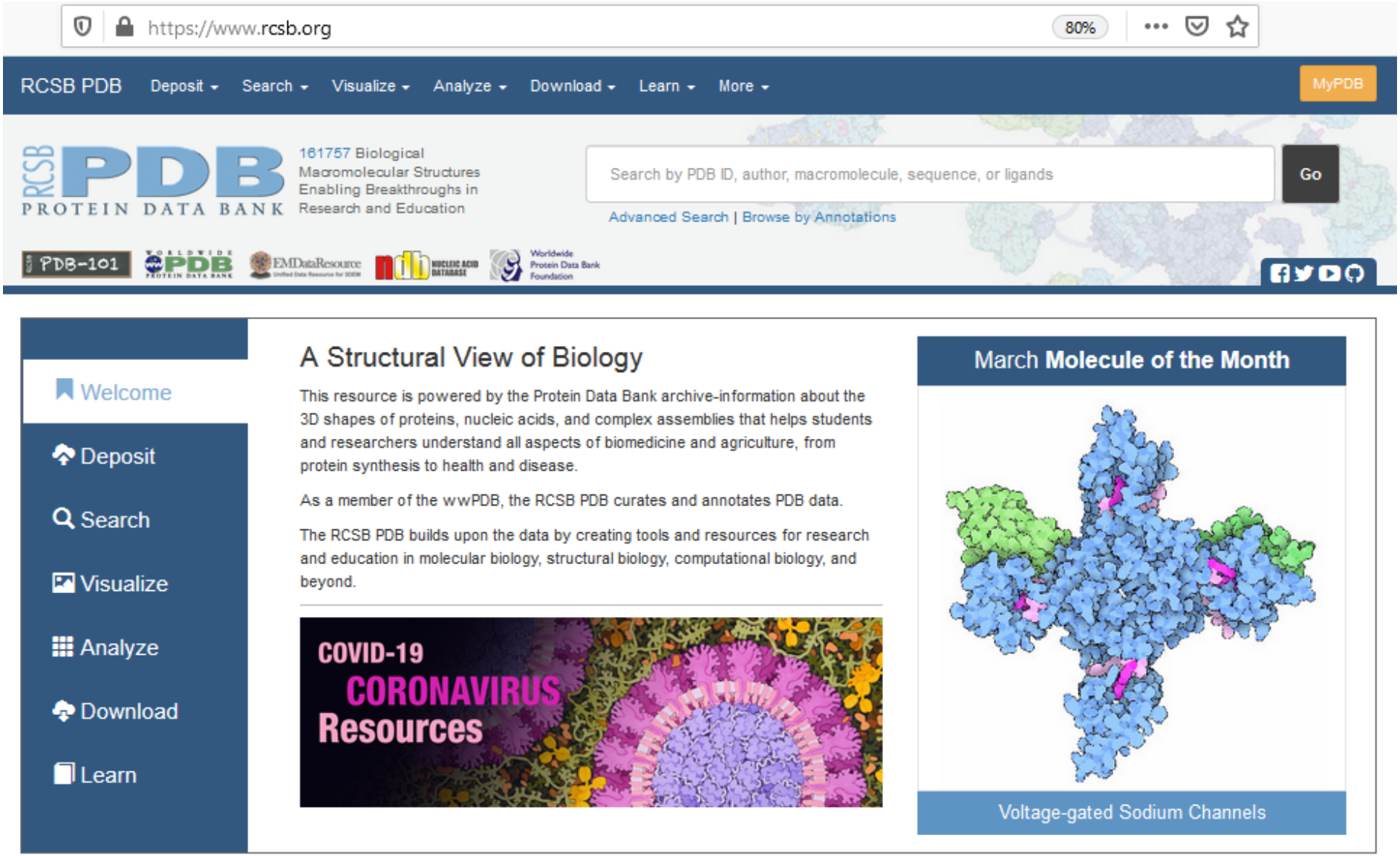
- Retrieve the abstract view (text format) of two PubMed records.
  - https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=pubmed&id=17284678,9997&retmode=text&rettype=abstract
- Retrieve two PubMed records in XML format.
  - https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=pubmed&id=11748933,11700088&retmode=xml

ESearch

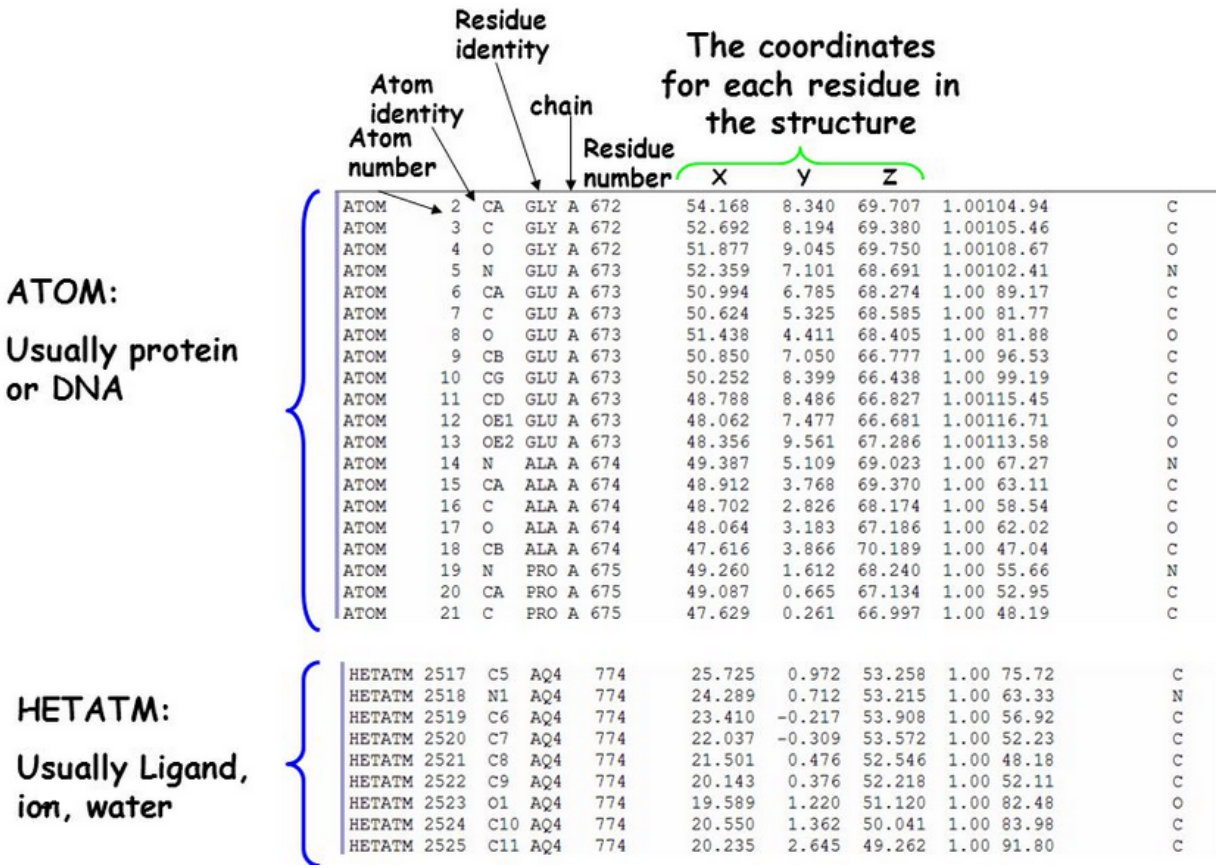
ESearch Formatting Parameters

- `retstart` (optional): Setting this parameter helps limit which of the unique identifiers (UIDs) in the results set will be shown in the output, as it determines whether the output begins at the first retrieved UID, or with a UID that is later in the results set. For example, if `retstart` is set to `10`, the first ten UIDs in the results set will be skipped, and the output will begin with the eleventh UID. The default of this parameter is `0`, corresponding to the first record in the entire set. This parameter can be used in conjunction with `retmax` to download an arbitrary subset of UIDs retrieved from a search.
- `retmax` (optional): Total number of unique identifiers (UIDs) from the retrieved set to be shown in the output (default=20). Increasing `retmax` allows more of the retrieved UIDs to be included in the output, up to a maximum of 100,000 UIDs. If you need to retrieve more than 100,000 UIDs, you can submit multiple ESearch requests, and increase the `retstart` parameter each time. For example:
  - https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?db=pubmed&term=cancer&retmax=100: This URL will return results 1 through 100 of a search for "Cancer".
  - https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?db=pubmed&term=cancer&retstart=100&retmax=100: This URL will return results 101 through 200 of the same search for "Cancer".
- `rettype` (optional): Retrieval type. There are two supported values:
  - `uilist` (default): Displays the standard XML output, including a list of unique identifiers (UIDs), the total number of results, and the query translation for the search.
  - `count`: Displays only the total number of results, without the list of UIDs or query translation.
- `retmode` (optional): Determines the format of the returned output. The default value is `xml`, but `json` is also supported.

PDB


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The RCSB PDB builds upon the data by creating tools and resources for research and education in molecular biology, structural biology, computational biology, and beyond.
COVID-19
CORONAVIRUS
Resources
March Molecule of the Month
Voltage-gated Sodium Channels

The PDB file – text format



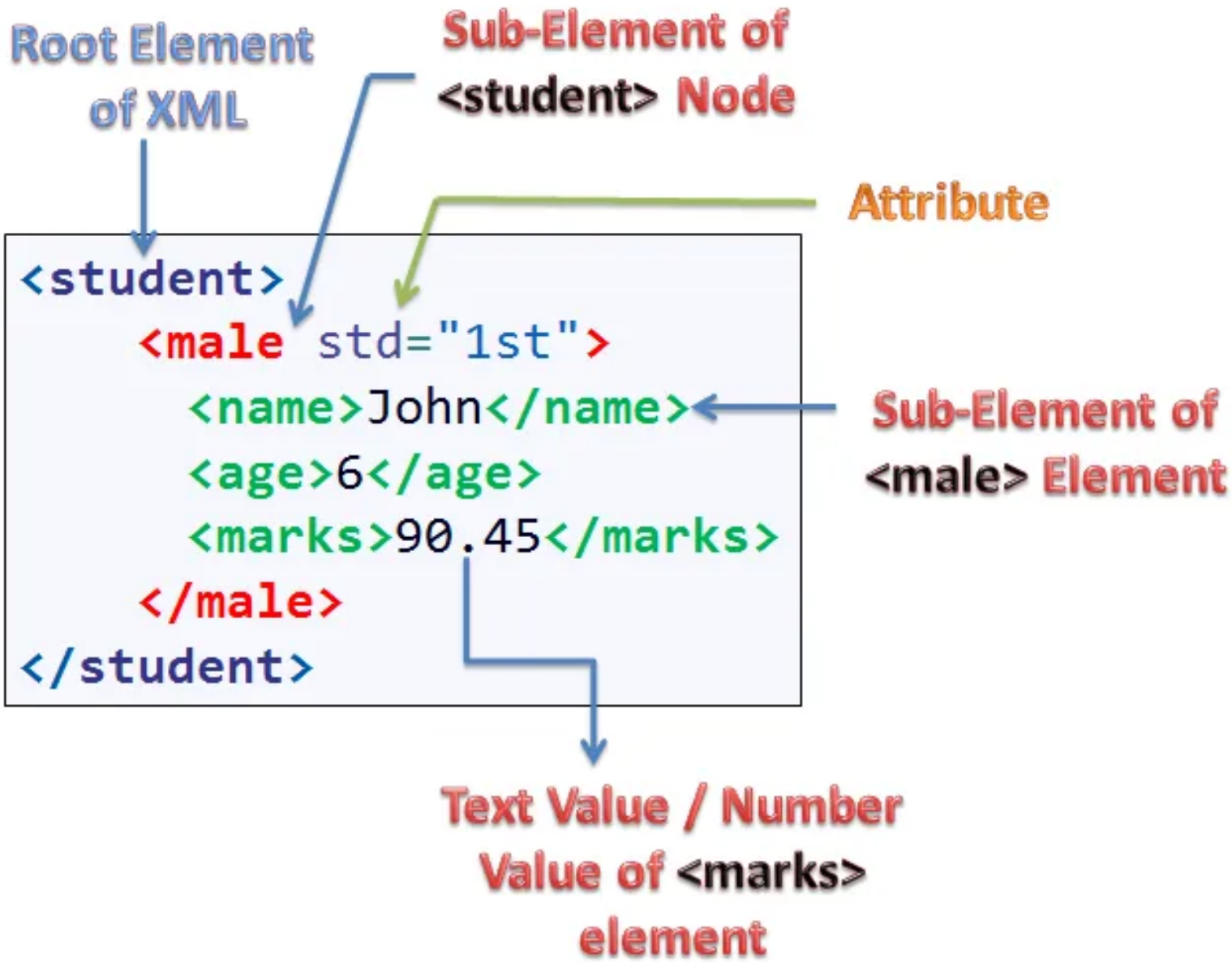

```
ATOM      2  CA  GLY A 672      54.168   8.340  69.707  1.00104.94           C
ATOM      3  C   GLY A 672      52.692   8.194  69.380  1.00105.46           C
ATOM      4  O   GLY A 672      51.877   9.045  69.750  1.00108.67           O
ATOM      5  N   GLU A 673      52.359   7.101  68.691  1.00102.41           N
ATOM      6  CA  GLU A 673      50.994   6.785  68.274  1.00 89.17           C
ATOM      7  C   GLU A 673      50.624   5.325  68.585  1.00 81.77           C
ATOM      8  O   GLU A 673      51.438   4.411  68.405  1.00 81.88           O
ATOM      9  CB  GLU A 673      50.850   7.050  66.777  1.00 96.53           C
ATOM     10  CG  GLU A 673      50.252   8.399  66.438  1.00 99.19           C
ATOM     11  CD  GLU A 673      48.788   8.486  66.827  1.00115.45           C
ATOM     12  OE1 GLU A 673      48.062   7.477  66.681  1.00116.71           O
ATOM     13  OE2 GLU A 673      48.356   9.561  67.286  1.00113.58           O
ATOM     14  N   ALA A 674      49.387   5.109  69.023  1.00 67.27           N
ATOM     15  CA  ALA A 674      48.912   3.768  69.370  1.00 63.11           C
ATOM     16  C   ALA A 674      48.702   2.826  68.174  1.00 58.54           C
ATOM     17  O   ALA A 674      48.064   3.183  67.186  1.00 62.02           O
ATOM     18  CB  ALA A 674      47.616   3.866  70.189  1.00 47.04           C
ATOM     19  N   PRO A 675      49.260   1.612  68.240  1.00 55.66           N
ATOM     20  CA  PRO A 675      49.087   0.665  67.134  1.00 52.95           C
ATOM     21  C   PRO A 675      47.629   0.261  66.997  1.00 48.19           C
```

```
HETATM 2517  C5  AQ4   774      25.725   0.972  53.258  1.00 75.72           C
HETATM 2518  N1  AQ4   774      24.289   0.712  53.215  1.00 63.33           N
HETATM 2519  C6  AQ4   774      23.410  -0.217  53.908  1.00 56.92           C
HETATM 2520  C7  AQ4   774      22.037  -0.309  53.572  1.00 52.23           C
HETATM 2521  C8  AQ4   774      21.501   0.476  52.546  1.00 48.18           C
HETATM 2522  C9  AQ4   774      20.143   0.376  52.218  1.00 52.11           C
HETATM 2523  O1  AQ4   774      19.589   1.220  51.120  1.00 82.48           O
HETATM 2524  C10 AQ4   774      20.550   1.362  50.041  1.00 83.98           C
HETATM 2525  C11 AQ4   774      20.235   2.645  49.262  1.00 91.80           C
```

XML

1. **XML** – Extensible Markup Language – rozszerzalny język znakowania przeznaczony do reprezentowania różnych danych w ustrukturalizowany sposób
    1. **znacznik** – ciąg liter, cyfr i innych znaków (bez spacji i zaczynający się od litery) ograniczony znakami $<$ i $>$. Tagowi (znacznikowi) otwierającemu zawsze towarzyszy tag zamykający zaczynający się od znaków $</$ i kończący się znakiem $>$.
    2. **atrybut** – definiowany w obrębie znacznika, składa się zwykle pary *nazwa* $=$ *wartość*, przy czym wartość zwykle podawana jest w cudzysłowach,
    3. **encja** – jest elementem treści dokumentu i reprezentuje pojedynczy symbol, encje definiuje się podając po znaku & numer symbolu i kończąc znakiem średnika, np & lt;

XML


Root Element of XML
Sub-Element of <student> Node
Attribute
<student>
<male std="1st">
<name>John</name>
<age>6</age>
<marks>90.45</marks>
</male>
</student>
Sub-Element of <male> Element
Text Value / Number Value of <marks> element

XML

```
<eInfoResult>
   <DbInfo>
      <DbName>pubmed</DbName>
      <MenuName>PubMed</MenuName>
      <Description>PubMed bibliographic record</Description>
      <Count>22595116</Count>
      <LastUpdate>2013/03/19 03:50</LastUpdate>
      <FieldList>
         <Field>
            <Name>ALL</Name>
            <FullName>All Fields</FullName>
            <Description>All terms from all searchable fields</Description>
            <TermCount>139835112</TermCount>
            <IsDate>N</IsDate>
            <IsNumerical>N</IsNumerical>
            <SingleToken>N</SingleToken>
            <Hierarchy>N</Hierarchy>
            <IsHidden>N</IsHidden>
         </Field>
         <Field>
         ...
</eInfoResult>
```

Wektory 2D - podstawowe operacje

Niech dane są $v = (1, 2)$ oraz $w = (4, 0)$. Wtedy:

1. **suma wekorów** v i w wynosi:
$$v + w = (1 + 4, 2 + 0) = (5, 2),$$
2. **iloczyn skalarny wektorów** v i w wynosi:
$$< v, w >= 1 \cdot 4 + 2 \cdot 0 = 4 + 0 = 4$$
3. **długość wektora** v wynosi:
$$||v|| = \sqrt{< v, v >} = \sqrt{1 \cdot 1 + 2 \cdot 2} = \sqrt{5}.$$

Wektory 3D - podstawowe operacje

Niech dane są $v = (1, 2, 3)$ oraz $w = (4, 0, -2)$. Wtedy:

1. **suma wekorów** v i w wynosi:
$$v + w = (1 + 4, 2 + 0, 3 + (-2)) = (5, 2, 1),$$
2. **iloczyn skalarny wektorów** v i w wynosi:
$$< v, w >= 1 \cdot 4 + 2 \cdot 0 + 3 \cdot (-2) = 4 + 0 - 6 = -2,$$
3. **długość wektora** v wynosi:
$$||v|| = \sqrt{< v, v >} = \sqrt{1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3} = \sqrt{14}.$$

Wektory ND - podstawowe operacje

Niech dane są $v = (v_1, v_2, ..., v_n)$ oraz $w = (w_1, w_2, ..., w_n)$. Wtedy:

1. **suma wekorów** v i w wynosi:
$$v + w = (v_1 + w_1, v_2 + w_2, ..., v_n + w_n),$$
2. **iloczyn skalarny wektorów** v i w wynosi:
$$< v, w >= v_1 w_1 + v_2 w_2 + ... + v_n w_n,$$
3. **długość wektora** v wynosi:
$$||v|| = \sqrt{< v, v >} = \sqrt{v_1 v_1 + v_2 v_2 + ... + v_n v_n} = \sqrt{v_1^2 + v_2^2 + ... + v_n^2}.$$

Przestrzenie wektorowe a analiza tekstu

Zdanie1: new york times
Zdanie2: new york post
Zdanie3: los angeles times

Przestrzenie wektorowe a analiza tekstu

Dokument1: new york times
Dokument2: new york post
Dokument3: los angeles times

Występujące słowa : angeles, los, new, post, times, york

Przestrzenie wektorowe a analiza tekstu

Dokument1: new york times
Dokument2: new york post
Dokument3: los angeles times

Występujące słowa (w ilu tekstach): angeles (1), los (1), new (2), post (1), times (2), york (2)

IDF (Inverse Document Frequency)

Liczba zdań $N = 3$.

Występujące słowa (w ilu tekstach): angeles (1), los (1), new (2), post (1), times (2), york (2)

$$idf_i = \log_2(\frac{N}{df_i})$$

gdzie df_i - liczba dokumentów zawierających i-te słowo. Wtedy:

- angles, $\log_2(\frac{3}{1}) = 1.584$,
- los, $\log_2(\frac{3}{1}) = 1.584$,
- new, $\log_2(\frac{3}{2}) = 0.584$,
- post, $\log_2(\frac{3}{1}) = 1.584$,
- times, $\log_2(\frac{3}{2}) = 0.584$,
- york, $\log_2(\frac{3}{2}) = 0.584$.

TF (Term Frequency)

Niech:

$$f_{ij} \text{ - częstość } i \text{ - tego słowa w dokumencie } j.$$

Wtedy:

$$tf_{ij} = \frac{f_{ij}}{\max\{f_{ij}\}}.$$

W naszym przypadku:

	angeles	los	new	post	times	york
d1	0	0	1	0	1	1
d2	0	0	1	1	0	1
d3	1	1	0	0	1	0

Ważenie TF-IDF

Definiujemy:

$$w_{ij} = tf_{ij} idf_i = tf_{ij} \log_2(\frac{N}{df_i}).$$

Uwaga:

- słowa występujące często w jakimś dokumencie, ale rzadko w pozostałych dokumentach będą miały wysoką wagę
- eksperymentalnie takie ważenie się sprawdza

Ważenie TF-IDF

Definiujemy:

$$w_{ij} = tf_{ij} idf_i = tf_{ij} \log_2(\frac{N}{df_i}).$$

W naszym przypadku:

	angeles	los	new	post	times	york
d1	0	0	0.584	0	0.584	0.584
d2	0	0	0.584	1.584	0	0.584
d3	1.584	1.584	0	0	0.584	0

Zapytanie

Powiedzmy, że mamy nowy dokument:

new new times

Pytanie: Który dokument najbardziej przypomina?

Zapytanie

Powiedzmy, że mamy nowy dokument:

new new times

Pytanie: Który dokument najbardziej przypomina? Należy obliczyć ważenie $TD - IDF$ dla zapytania.

Zapytanie

Powiedzmy, że mamy nowy dokument:

new new times

Pytanie: Który dokument najbardziej przypomina? Należy obliczyć ważenie *TD* – *IDF* dla zapytania.

q	0	0	(2/2)*0.584=0.584	0	(1/2)*0.584=0.292	0

Wyznaczanie długości wektorów

Mamy:

- $||d1|| = \sqrt{0.584^2 + 0.584^2 + 0.584^2} = 1.011,$
- $||d2|| = \sqrt{0.584^2 + 1.584^2 + 0.584^2} = 1.786,$
- $||d3|| = \sqrt{1.584^2 + 1.584^2 + 0.584^2} = 2.316,$
- $||q|| = \sqrt{0.584^2 + 0.292^2} = 0.652.$

Podobieństwo pomiędzy wektorami

Mamy:

$$\cos(d, q) = \frac{<d,q>}{||d|| \cdot ||q||}$$

Zadanie:
Wyznaczy miarę kosinus pomiędzy sekwencjami $d1$, $d2$, $d3$ a q.

Podobieństwo pomiędzy wektorami

Mamy:

$$\cos(d, q) = \frac{<d,q>}{||d|| \cdot ||q||}$$

- $\cos(d1, q) = \frac{0 \cdot 0 + 0 \cdot 0 + 0.584 \cdot 0.584 + 0 \cdot 0 + 0.584 \cdot 0.292 + 0.584 \cdot 0}{1.011 \cdot 0.652} = 0.776$
- $\cos(d2, q) = \frac{0 \cdot 0 + 0 \cdot 0 + 0.584 \cdot 0.584 + 1.584 \cdot 0 + 0 \cdot 0.292 + 0.584 \cdot 0}{1.786 \cdot 0.652} = 0.292$
- $\cos(d3, q) = \frac{1.584 \cdot 0 + 1.584 \cdot 0 + 0 \cdot 0.584 + 0 \cdot 0 + 0.584 \cdot 0.292 + 0 \cdot 0}{2.316 \cdot 0.652} = 0.112$

W jakiej kolejności należałoby zwrócić dokumenty na podane zapytanie?